

Seminar On Algebraic Geometry
Lecture Notes 2 (Incomplete version)

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In this lecture, we will discuss sheaves of abelian groups (moreover, sheaves of modules) by viewing Sh as an abelian category.

1 Abelian Category Of Sheaves

Let X be a topological space.

Definition 1.1. Let $\mathscr{F}'$ be a subsheaf of a sheaf $\mathscr{F}$, i.e. there is an injective morphism $i : \mathscr{F}' \to \mathscr{F}$ (we can view $\mathscr{F}'(U)$ as a subsheaf of $\mathscr{F}(U)$ for each open subset U). We define the quotient sheaf $\mathscr{F}/\mathscr{F}'$ to be the sheafification of the presheaf $U \mapsto \mathscr{F}(U)/\mathscr{F}'(U)$. For any $P \in X$, we have $(\mathscr{F}/\mathscr{F}')_P = \mathscr{F}_P/\mathscr{F}'_P$.

Corollary 1.2. [Ha, Exercise II.1.7] Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then $\operatorname{im}\varphi \cong \mathscr{F}/\ker\varphi$, and $\operatorname{coker}\varphi \cong \mathscr{G}/\operatorname{im}\varphi$.

Proposition 1.3 (Direct sum)**.** [Ha, Exercise II.1.9]

Proposition 1.4 (Direct limit)**.** [Ha, Exercise II.1.10, 1.11]

Proposition 1.5 (Inverse limit)**.** [Ha, Exercise II.1.12]

2 Operations on Sheaves

2.1 Global Section Functor

Recall that we sometimes write $\Gamma(U, \mathscr{F}) := \mathscr{F}(U)$, for any open subset $U \subseteq X$ and any sheaf $\mathscr{F}$ on X.

Proposition 2.1. [Ha, Ex 1.8] Suppose $U \subseteq X$ is an open subset, and $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is an exact sequence of sheaves of abelian groups. Then

$$0 \to \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G}) \to \Gamma(U, \mathscr{H})$$

is exact. I.e., $\Gamma(U, -)$ is a left exact functor $\mathrm{Sh}(X) \to \mathrm{Ab}$.

Proof. We can easily check this on the level of stalks, since for any morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves, and for any $P \in U$, we have $\ker\varphi_P = (\ker\varphi)_P$ and $\operatorname{im}\varphi_P = (\operatorname{im}\varphi)_P$. □

However, $\Gamma(U, -)$ is not exact in general, especially in algebraic geometry: a surjective morphism of sheaves need not induce a surjection on sections. The following two examples illustrate that differential topology is "soft," whereas the Zariski topology is "hard."

Lemma 2.2. Let M be a smooth (paracompact) manifold and let $\{U_i\}_{i\in I}$ be an open cover of M. Then there exists a locally finite partition of unity subordinate to $\{U_i\}$: a family $\{\rho_i\}_{i\in I}$ of smooth functions $\rho_i \in C^\infty(M)$ such that $0 \le \rho_i \le 1$, $\operatorname{supp}\rho_i \subseteq U_i$ is compact, the family $\{\operatorname{supp}\rho_i\}_{i\in I}$ is locally finite, and $\sum_{i\in I}\rho_i = 1$ on M (the sum is locally finite, hence well defined).

Example 2.3. Let M be a smooth manifold, $\mathscr{C}^\infty_M$ the sheaf of smooth functions on M, and $\mathscr{C}^\infty_{M,c}$ the presheaf of compactly supported smooth functions. The inclusion $\mathscr{C}^\infty_{M,c} \hookrightarrow \mathscr{C}^\infty_M$ induces an isomorphism

$$(\mathscr{C}^\infty_{M,c})^{\mathrm{sh}} \xrightarrow{\sim} \mathscr{C}^\infty_M$$

on sheafification. Indeed, for each $f \in C^\infty(M)$ and each point, choose a locally finite open cover $\{U_i\}_{i\in I}$ of M by relatively compact open subsets and a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to it. Then each $f_i := \rho_i f$ is compactly supported and only finitely many f_i are nonzero near each point, so $f = \sum_i f_i$ with $\sum_i f_i \in \Gamma(M, \mathscr{C}^\infty_{M,c})$. Thus every smooth function is locally a sum of compactly supported smooth functions, and the sheafification of $\mathscr{C}^\infty_{M,c}$ is $\mathscr{C}^\infty_M$.

Proposition 2.4. Let M be a smooth manifold and $\mathscr{O}_M := \mathscr{C}^\infty_M$ the sheaf of smooth functions. Then the global section functor

$$\Gamma(M, -) : \mathscr{O}_M\text{-}\mathsf{Mod} \longrightarrow \mathsf{Ab}$$

is exact.

Proof. Since $\Gamma(M, -)$ is left exact, it suffices to show that every surjective morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of $\mathscr{O}_M$-modules induces a surjection on global sections. Let $g \in \Gamma(M, \mathscr{G})$. As φ is an epimorphism of sheaves, there exist an open cover $\{U_i\}_{i\in I}$ of M and local lifts $f_i \in \mathscr{F}(U_i)$ with $\varphi(f_i) = g|_{U_i}$. Choose a partition of unity $\{\rho_i\}_{i\in I}$ subordinate to $\{U_i\}_{i\in I}$. Since $\mathscr{F}$ is an $\mathscr{O}_M$-module, the section $\rho_i f_i \in \mathscr{F}(U_i)$ vanishes near the boundary of U_i, hence extends by zero to a global section $\widetilde{\rho_i f_i} \in \Gamma(M, \mathscr{F})$. By local

finiteness the sum $f := \sum_{i\in I} \widetilde{\rho_i f_i}$ is a well-defined global section of $\mathscr{F}$, and

$$\varphi(f) = \sum_{i\in I} \rho_i\, \varphi(f_i) = \sum_{i\in I} \rho_i\, g|_{U_i} = g.$$

Hence φ is surjective on global sections, and $\Gamma(M, -)$ is exact. □

Example 2.5. [Va, Example 2.4.10] Let $X = \mathbb{C}$ with the classical topology, and define $\mathscr{O}_X$ to be the sheaf of holomorphic functions, and $\mathscr{O}_X^*$ to be the sheaf of invertible holomorphic functions. Then we have an exact sequence

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}_X^* \longrightarrow 1,$$

where $\underline{\mathbb{Z}}$ is the constant sheaf associated to $\mathbb{Z}$.

Indeed, we can check this on stalks. Without loss of generality, consider $P = 0/$ Then $\underline{\mathbb{Z}}_0 = \mathbb{Z}$, and $\mathscr{O}_{X,0}$ is the ring of convergent Taylor series at origin, and $\mathscr{O}_{X,0}^*$ is the ring of functions in $\mathscr{O}_{X,0}$ which doesn't vanish at 0. The image of $\times 2\pi i$ on stalks is $2\pi i\mathbb{Z}$ as constant functions, and the kernel of exp is just $2\pi i\mathbb{Z}$ by Euler's equation. To show that the expotential map is surjective on stalks, we only need to notice that for any holomorphic function $f \in \mathscr{O}_{X,0}^*$, there is an open neighborhood U of the origin such that f has no roots in U. Thus, $\ln f$ is a holomorphic function in U, which gives the preimage of f in $\mathscr{O}_{X,0}$.

Let $U = \mathbb{C} \setminus \{0\}$ be the punctured complex plane. Suppose that there is some $f \in \mathscr{O}_X(U)$, such that $\exp f = \mathrm{id}_U \in \mathscr{O}_X^*(U)$, i.e., $\exp(f(z)) = z$ on U. Then $f'(z)z = f'(z)e^{f(z)} = 1$, for $z \in U$, and we have $f'(z) = 1/z$. Now we calculate the integral at $\gamma : |z| = 1$:

$$\oint_\gamma f'(z)\,\mathrm{d}z = \oint_\gamma \frac{1}{z}\,\mathrm{d}z = 2\pi i \neq 0.$$

However, by Cauchy's Integral Theorem, since γ is a closed curve containing in U, a holomorphic open domain of $f'(z)$, the integral must be zero. Therefore, such f does not exist, and $\exp(U)$ is not surjective.

2.2 Hom Sheaf

Proposition 2.6. [Ha, Exercise II.1.15] Let $\mathscr{F}, \mathscr{G}$ be sheaves of abelian groups on X. For any open set $U \subseteq X$, the set $\mathscr{H}om\,(\mathscr{F}\,|_U, \mathscr{G}\,|_U)$ is an abelian group, and $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is a sheaf of abelian groups on X, by $U \mapsto \mathrm{Hom}\,(\mathscr{F}\,|_U, \mathscr{G}\,|_U$). It is called the *sheaf of local morphisms* of $\mathscr{F}$ into $\mathscr{G}$, "sheaf hom" for short,

and is denoted $\mathrm{Hom}\,(\mathscr{F},\mathscr{G})$.

Proof. For each open $U \subseteq X$, the set $\mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ becomes an abelian group under pointwise addition: for morphisms φ, ψ and any open $V \subseteq U$, $s \in \mathscr{F}(V)$, put

$$(\varphi + \psi)(V)(s) = \varphi(V)(s) + \psi(V)(s),$$

and define $(-\varphi)(V)(s) = -\varphi(V)(s)$, the zero morphism sending every section to 0. The group axioms are inherited from the abelian groups $\mathscr{G}(V)$, so $U \mapsto \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ is a presheaf of abelian groups.

It remains to check the sheaf axioms. Let $U \subseteq X$ be open and $\{U_i\}_{i\in I}$ an open cover of U. *Identity:* if $\varphi \in \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ satisfies $\varphi|_{U_i} = 0$ for all i, then for every open $V \subseteq U$ and $s \in \mathscr{F}(V)$ the sections $\varphi(V)(s)|_{V\cap U_i} = 0$ for all i; since $\mathscr{G}$ is a sheaf and $\{V \cap U_i\}_{i\in I}$ covers V, we get $\varphi(V)(s) = 0$, hence $\varphi = 0$. *Gluing:* suppose $\varphi_i \in \mathrm{Hom}(\mathscr{F}\mid_{U_i}, \mathscr{G}\mid_{U_i})$ satisfy $\varphi_i|_{U_i\cap U_j} = \varphi_j|_{U_i\cap U_j}$ for all i, j. For open $V \subseteq U$ and $s \in \mathscr{F}(V)$, the sections $\varphi_i(V \cap U_i)(s|_{V\cap U_i}) \in \mathscr{G}(V \cap U_i)$ agree on the overlaps $V \cap U_i \cap U_j$; by the sheaf property of $\mathscr{G}$ they glue to a unique section $\varphi(V)(s) \in \mathscr{G}(V)$. The assignments $V \mapsto \varphi(V)$ define a morphism of sheaves $\mathscr{F}\mid_U \to \mathscr{G}\mid_U$, and $\varphi|_{U_i} = \varphi_i$ by construction. Hence the presheaf is a sheaf of abelian groups. □

Proposition 2.7. [Va, Exercise 2.6.I] Suppose $\mathscr{F}$ is a sheaf of abelian groups on a topological space X. Then $\mathscr{H}om\,(\mathscr{F},\cdot)$ is a left-exact functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, a nd $\mathscr{H}om\,(\cdot,\mathscr{F})$ is a left-exact contravariant functor $\mathrm{Sh}(X) \to \mathrm{Sh}(X)$, as we define above.

Proof. Since the Hom sheaf is defined on sections by $\mathscr{H}om(\mathscr{F},\mathscr{G})(U) = \mathrm{Hom}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$, and exactness of a sequence of sheaves is checked on every open set (restriction to an open subset preserves exactness), it suffices to prove the corresponding exactness of Hom in $\mathrm{Sh}(U)$ for each open U.

Let $0 \to \mathscr{G}' \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{G}''$ be exact. We show that

$$0 \to \mathrm{Hom}(\mathscr{F},\mathscr{G}') \xrightarrow{\alpha_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}) \xrightarrow{\beta_*} \mathrm{Hom}(\mathscr{F},\mathscr{G}'')$$

is exact. If $\alpha \circ \varphi = 0$, then $\varphi = 0$ since α is injective; hence α_* is injective. As $\beta \circ \alpha = 0$, we have $\beta_* \circ \alpha_* = 0$. Conversely, if $\beta \circ \varphi = 0$, then φ factors through $\ker\beta = \mathrm{im}\,\alpha \cong \mathscr{G}'$ (using that α is injective), so $\varphi = \alpha \circ \psi$ for a unique $\psi \in \mathrm{Hom}(\mathscr{F},\mathscr{G}')$. Thus $\ker\beta_* = \mathrm{im}\,\alpha_*$, proving left-exactness of $\mathscr{H}om(\mathscr{F},\cdot)$. □

3 Pushforward & Inverse Image

Definition 3.1. Let $f : X \to Y$ be a continuous map of topological spaces.

- For any sheaf $\mathscr{F}$ on X, we define the *direct image (pushforward)* sheaf $f_*\mathscr{F}$ on Y by
$$f_*\mathscr{F}(V) := \mathscr{F}(f^{-1}V),$$
for any open set $V \subseteq Y$.
- For any sheaf $\mathscr{G}$ on Y, we define a presheaf $f^+\mathscr{G}$ on X by
$$f^+\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V),$$
for any open set $U \subseteq X$. And we define *inverse image* sheaf $f^{-1}\mathscr{G}$ to be the sheafification of $f^+\mathscr{G}$ given above.

Then $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ and $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ are two functors between sheaves on different topological spaces.

It's easy to verify that the pushforward $f_*\mathscr{F}$ is still a sheaf, since for any open subset $V \subseteq Y$, $f^{-1}V$ is an open subset of X. However, let $U \subseteq X$ be an open subset, it's not in general $f(V)$ is still open. Therefore, we take an approximation by the larger open sets $V \supseteq f(U)$. More precisely, since the inclusion determines a reversed restriction map, we define the inverse image to be the limit of such $\mathscr{G}(V)$, with arrows $\mathscr{G}(V) \to f^{-1}\mathscr{G}(U)$.

If $x \in X$ is a point, and $i : \{x\} \hookrightarrow X$ is the inclusion map, then the stalk can be viewed as $\mathscr{F}_x = i^{-1}\mathscr{F}$ for every presheaf $\mathscr{F}$ on X.

Lemma 3.2. [Va, Exercise 2.2.I, 2.7.C] Suppose $f : X \to Y$ is a continuous map, and $\mathscr{F}$ is a sheaf on X, $\mathscr{G}$ is a sheaf on Y. If $f(p) = q$, then we have
$$(f_*\mathscr{F})_q \cong \mathscr{F}_p, \text{ and } \mathscr{G}_q \cong (f^{-1}\mathscr{G})_p.$$

Proof. We prove this statement by the definition of colimit. On the one hand, for each open neighborhood U of q, $f^{-1}U$ is an open neighborhood of p, thus there is a morphism $\mathscr{F}_p \to \varinjlim_{q \in U} \mathscr{F}(f^{-1}U) = (f_*\mathscr{F})_q$; on the other hand, for each open neighborhood V of p, $V \subseteq f^{-1}f(V)$ and $q \in f(V)$, which shows that $\mathscr{F}_p \cong (f_*\mathscr{F})_q$, since the limit is given by the inclusions of open subsets.

The analogue of inverse image is given by the equation

$$(f^{-1}\mathscr{G})_p = (f^+\mathscr{G})_p = \varinjlim_{p\in U} \varinjlim_{f(U)\subseteq V} \mathscr{G}(V) = \varinjlim_{q\in V} \mathscr{G}(V),$$

where the last equation is because for each open neighborhood V of q, $q \in f^{-1}(V)$ and $ff^{-1}(V) \subseteq V$. □

Proposition 3.3. [Ha, Exercise II.1.18] Let $f : X \to Y$ be a continuous map of topological space. Then $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is left-adjoint to $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$. Precisely, there is a natural isomorphism

$$\tau_{\mathscr{F}\mathscr{G}} : \mathrm{Hom}_X\left(f^{-1}\mathscr{G}, \mathscr{F}\right) \to \mathrm{Hom}_Y\left(\mathscr{G}, f_*\mathscr{F}\right)$$

for any sheaf $\mathscr{F} \in \mathrm{Sh}(X)$ and $\mathscr{G} \in \mathrm{Sh}(Y)$.

Proof. We prove the adjointness by giving the unit and counit. Notice that for open subset $V \subseteq Y$, there is a natural morphism

$$\eta_{\mathscr{G}}(V) : \mathscr{G}(V) \to f_*f^+\mathscr{G}(V) = \varinjlim_{V'\supseteq f(f^{-1}V)} \mathscr{G}(V') = \varinjlim_{V'\supseteq V\cap f(X)} \mathscr{G}(V'),$$

since $V \supseteq V \cap f(X)$. On the other side, for open subset $U \subseteq X$, there is also a natural morphism

$$\varepsilon_{\mathscr{F}}(U) : f^+f_*\mathscr{F}(U) = \varinjlim_{U'\supseteq f(U)} \mathscr{F}(f^{-1}U') \to \mathscr{F}(U),$$

since U is contained in each $f^{-1}U'$. □

Definition 3.4. If Z is a subset of X, regarded as a topological subspace with the induced topology , if $i : Z \to X$ is the inclusion map, and if $\mathscr{F}$ is a sheaf on X, then we call $i^{-1}\mathscr{F}$ the *restriction* of $\mathscr{F}$ to Z, and we often denote it by $\mathscr{F}\mid_Z$. If U is an open subset of X, then $\mathscr{F}\mid_U$ is naturally defined by $\mathscr{F}\mid_U(V) = \mathscr{F}(V)$ for any open subset $V \subseteq U$.

Corollary 3.5. Let Z be a subset of X, $\mathscr{F}$ be a sheaf on X. Then $(\mathscr{F}\mid_Z)_P = \mathscr{F}_P$ for any point $P \in Z$.

Example 3.6 (Skyscraper)**.** [Ha, Exercise II.1.17]

4 Locally Ringes Spaces

Lemma 4.1. Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be two local rings, and $\varphi : A \to B$ be a homomorphism, then the following statements are equivalent:

(a) $f^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$,

(b) $f(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

If f satifies the equivalent condition above, we say that f is a **local homomorphism**.

Proof. (a) $\implies$ (b) is because $f(\mathfrak{m}_A) = ff^{-1}(\mathfrak{m}_B) = \mathfrak{m}_B \cap f(A) \subseteq \mathfrak{m}_B$. Suppose $f(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$, then $\mathfrak{m}_A \subseteq f^{-1}f\mathfrak{m}_A \subseteq f^{-1}(\mathfrak{m}_B)$. Since $\mathfrak{m}_A$ is maximal, and $f^{-1}(\mathfrak{m}_B)$ is a prime ideal, we have $f^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$, this proves (b) $\implies$ (a). □

Definition 4.2. A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ on X.

A **morphism of ringed spaces** from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ of a continuous map $f : X \to Y$ and a morphism of sheaves $\mathscr{O}_Y \to f_*\mathscr{O}_X$ on Y.

We say the ringed space $(X, \mathscr{O}_X)$ is a **locally ringed space**, if for each point $P \in X$, the stalk $\mathscr{O}_{X,P}$ is a local ring, with a unique maximal ideal $\mathfrak{m}_{X,P}$.

A **morphism of locally ringed spaces** is a morphism $(f, f^\sharp)$ of ringed spaces, such that for each point $P \in X$, the induced natural map

$$f_P^\sharp : \mathscr{O}_{Y,f(P)} = \varinjlim_{f(P)\in V} \mathscr{O}_Y(V) \to \varinjlim_{f(P)\in V} \mathscr{O}_X(f^{-1}V) \to \mathscr{O}_{X,P}$$

is a local homomorphism.

Notice that, by Lemma 3.2, the induced map on stalks is given by

$$f_P^\sharp : \mathscr{O}_{Y,f(P)} \to (f_*\mathscr{O}_X)_{f(P)} \cong \mathscr{O}_{X,P}.$$

Next lemma show the equivalent definition of the local induced map.

Here is a geometric view, to view a section $f \in \mathscr{O}_X(U)$ as a function. It takes values at $P \in X$ by $g(P) := g_P + \mathfrak{m}_P \in \mathscr{O}_{X,P}/\mathfrak{m}_{X,P}$. So we can restate the condition of local homomorphism by the following lemma.

Lemma 4.3. [E&H] Let $(f, f^\sharp) : X \to Y$ be a morphism of ringed spaces, where X and Y are locally ringed spaces. Then the following statements are equivalent:

(a) f is a morphism of locally ringed spaces, i.e., the induced map $f_P^\sharp : \mathscr{O}_{Y,f(P)} \to \mathscr{O}_{X,P}$ is a local homomorphism for any $P \in X$.

(b) For any point $P \in X$ and any open neighborhood V of $f(P)$ in Y, a section $s \in \mathscr{O}_Y(V)$ vanishes at $f(P)$, i.e., $s_{f(P)} \in \mathfrak{m}_{Y,f(P)}$ if and only if the section $f^\sharp(s) \in f_*\mathscr{O}_X(V) = \mathscr{O}_X(f^{-1}V)$ vanishes at P, i.e., $f^\sharp(s)_P \in \mathfrak{m}_{X,P}$.

Proof. (a) $\implies$ (b): Let $s \in \mathscr{O}_Y(V)$ be a section. Suppose that $s_{f(P)} \in \mathfrak{m}_{Y,f(P)}$. Then $(f^\sharp(s))_P = f_P^\sharp(s_{f(P)}) \subseteq \mathfrak{m}_{X,P}$, by Lemma 4.1. Conversely, suppose that $f^\sharp(s)_P \in \mathfrak{m}_{X,P}$. Then $s_{f(P)} \in f_P^{\sharp,-1}(f^\sharp(s)_P) = \mathfrak{m}_{Y,f(P)}$.

(b) $\implies$ (a): Now consider a germ $\langle s, V\rangle$, where $s \in \mathscr{O}_Y(V)$ and V is an open neighborhood of $f(P)$. Suppose that $\langle s, V\rangle \in \mathfrak{m}_{Y,f(P)}$. Then $f^\sharp(s)_P = f_P^\sharp(\langle s, V\rangle) = \langle f^\sharp(s), f^{-1}V\rangle = f^\sharp(s)_P \in \mathfrak{m}_{X,P}$. This proves that $f_P^\sharp(\mathfrak{m}_{Y,f(P)}) \subseteq \mathfrak{m}_{X,P}$, thus $f^\sharp$ is a local homomorphism. □

Example 4.4. Let $(X, \mathscr{C}_X)$ be a ringed space, where X is a topological space and $\mathscr{C}_X$ denotes the sheaf of continuous maps from X to $\mathbb{R}$ (or $\mathbb{C}$). Let $f : X \to Y$ be a continuous map, then there is a pullback

$$f^\sharp : \mathscr{C}_Y(U) \to f_*\mathscr{C}_X(U) = \mathscr{C}_X(f^{-1}U), \quad g \mapsto g \circ f$$

for any open subset $U \subseteq Y$, which gives the morphism of sheaves $f^\sharp : \mathscr{C}_Y \to f_*\mathscr{C}_X$.

Now consider the stalks. Since there continuous function $g : X \to \mathbb{R}$ can be defined locally, $\mathscr{C}_{X,P}$ is simply a $\mathbb{R}$-algebra of functions continuous at $P \in X$. $\mathscr{C}_{X,P}$ is a local ring, with a unique maximal ideal $\mathfrak{m}_{X,P} = \{h \in \mathscr{C}_{X,P} : h(P) \neq 0\}$ (because if f does not vanish at P, then there is an open neighborhood of P such that f does not vanish on it), which is the kernel of the value map $\mathfrak{m}_{X,P} \to \mathbb{R}$ given by $h \mapsto h(P)$. If $g \in \mathscr{C}_{Y,f(P)}$ is a continuous function at $f(P)$, we have $f^\sharp(g)(f(P)) = g \circ f(P) = 0$ if and only if $g(f(P)) = 0$. Therefore, $f_P^\sharp$ is a local homomorphism, by 4.3.

Example 4.5. Let $(M, \mathscr{C}_M^\infty)$ be a ringed space, where M is a smooth manifold (as a topological manifold), $\mathscr{C}_M^\infty$ is the ring of smooth functions on M.

Let $(N, \mathscr{C}_N^\infty)$ be another ringed space of smooth manifold, $f : M \to N$ a continuous map. Then we can also define the pullback map $f^\sharp : \mathscr{C}_N^\infty \to f_*\mathscr{C}_M$.

Indeed, [Wed, Proposition 4.18] shows that: $f : M \to N$ is a smooth map, if and only if the pullback map $f^\sharp$ factors through $f_*\mathscr{C}_M^\infty$, i.e., the pullback of each smooth function on N is a smooth function on M. (In [Wed], the author provides a view to construct a differential manifold totally as a ringed space.) Therefore, $f^\sharp : \mathscr{C}_N^\infty \to f_*\mathscr{C}_M^\infty$ is well-defined, and the induced map $f_P^\sharp : \mathscr{C}_{Y,f(P)} \to f_*\mathscr{C}_{X,P}$ is a local homomorphism by viewing the similar arguments with the continuous version above.

Similarly, complex manifold is also a locally ringed space.

Indeed, the equivalent definition of manifolds motivate the definition of morphisms between algebraic varieties: a continuous map $f : X \to Y$ between varieties is a morphism if it pull back a regular function on a $V \subseteq Y$ to a regular function on $f^{-1}(V) \subseteq X$.

Example 4.6. Let $(X, \mathscr{O}_X)$ be a variety, with the regular function sheaf as the structure sheaf. By [Ha, Theorem 3.2, 3.4], the stalks $\mathscr{O}_{X,P}$ is a local ring (localization of the coordinate ring at the maximal ideal corresponding to P). The morphism $f : X \to Y$ of varieties is a continuous map such that $f^* : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is well-defined by sending a regular function $s \in \mathscr{O}_Y(V)$ to the pullback $s \circ f \in \mathscr{O}_X(f^{-1}V)$.

Moreover, since for each maximal ideal $\mathfrak{m}_{X,P}$ in the local ring $\mathscr{O}_{X,P}$, we have $\mathscr{O}_{X,P}/\mathfrak{m}_{X,P} \cong k$ via the valuation map $f \mapsto f(P)$. It's easy to verify that the induced map $f_P^\sharp : \mathscr{O}_{Y,f(P)} \to \mathscr{O}_{X,P}$ is a local homomorphism.

5 Sheaves Of Modules: An Introduction

6 Base Sheaves

This section is based on [E&H, Proposition I-12], [Va, 2.5 Recovering sheaves from a "sheaf on a base"] and [Sta, 6.30 Bases and sheaves].

Definition 6.1. Let X be a topological space. Let $\mathcal{B}$ be a basis for X. We say that a collection of groups $\mathscr{F}(U)$ for open sets $U \in \mathcal{B}$ and maps $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ for $V \subseteq U$ form a $\mathcal{B}$-*sheaf* if they satisfy the sheaf axiom with respect to inclusions of basic open sets in basic open sets and coverings of basic open sets by basic open sets:

- **Presheaf axiom:** $\mathscr{F}$ is a contravarient functor from the category of $\mathcal{B}$ with morphisms of inclusions, to the category of Ab.
- **Sheaf axiom:** For any $U \in \mathcal{B}$ and an open cover $\{U_i\}_{i\in I} \subseteq \mathcal{B}$ of U, if there is a collection of sections $s_i \in \mathscr{F}(U_i)$, such that for any covers U_i and U_j, and any basic open sets $V \in \mathcal{B}$ such that $V \subseteq U_i \cap U_j$, $s_i \mid_V = s_j \mid_V$, then there is a unique section $s \in \mathscr{F}(U)$, such that $s \mid_{U_i} = s_i$.

A morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of $\mathcal{B}$-sheaves is a set of morphisms $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$, for any $U \in \mathcal{B}$, such that $\varphi(s \mid_V) = \varphi(s) \mid_V$, if $V \subseteq U$ and $s \in \mathscr{F}(U)$.

Lemma 6.2. Let $\mathcal{B}$ be a base of open sets for X, $\mathscr{F}$ be a $\mathcal{B}$-sheaf. For any open set $U \subseteq X$, define $\mathscr{F}(U)$ as the inverse limit of the sets $\mathscr{F}'(V)$, where V runs over basic open sets contained in U:

$$\mathscr{F}'(U) = \varprojlim_{V\subseteq U, V\in\mathcal{B}} \mathscr{F}(V).$$

Then

- $\mathscr{F}'(U)$ is the set of families $(f_V)_{V\subseteq U, V\in\mathcal{B}} \in \prod_{V\subseteq U, V\in\mathcal{B}} \mathscr{F}(V)$ such that $f_V \mid_W = f_W$, whenever $W \subseteq V \subseteq U$ with $W, V \in \mathcal{B}$.
- $\mathscr{F}'(V) = \mathscr{F}(V)$ for all $V \in \mathcal{B}$.
- $\mathscr{F}'$ is a sheaf on X.

Proposition 6.3. [E&H, Proposition I-12] Let $\mathcal{B}$ be a base of open sets for X.

(1) Every $\mathcal{B}$-sheaf $\mathscr{F}$ on X extends uniquely to a sheaf on X, denoted by $\mathscr{F}^{\text{ext}} \in \mathsf{Sh}(X)$.

(2) Every morphism of $\mathcal{B}$-sheaves $\overline{\varphi} : \mathscr{F} \to \mathscr{G}$ extends uniquely to a morphism $\varphi : \mathscr{F}^{\text{ext}} \to \mathscr{G}^{\text{ext}}$ of sheaves on X, such that $\varphi(U) = \overline{\varphi}(U)$, for all $U \in \mathcal{B}$.

The extension above forms a functor $\text{ext} : \text{Sh}(\mathcal{B}) \to \text{Sh}(X)$, from the category of $\mathcal{B}$-sheaf to the category of sheaves on X.

7 Gluing Sheaves And Locally Ringed Spaces

Proposition 7.1. [Ha, Exercise II.1.22] Let X be a topological space, let $\mathcal{U} = \{U_i\}$ be an open cover of X, and suppose we are given for each i a sheaf $\mathscr{F}_i$ on U_i, and for each i, j, an isomorphism $\varphi_{ij} : \mathscr{F}_i \mid_{U_i \cap U_j} \to \mathscr{F}_j \mid_{U_i \cap U_j}$, such that

(1) for each i, $\varphi_{ii} = \mathrm{id}$, and

(2) for each i, j, k, $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_i \cap U_j \cap U_k$.

Then there exists a unique sheaf $\mathscr{F}$ on X, together with isomorphisms $\psi_i : \mathscr{F} \mid_{U_i} \to \mathscr{F}_i$ such that for each i, j, $\psi_j = \varphi_{ij} \circ \psi_i$ on $U_i \cap U_j$. We say that $\mathscr{F}$ is obtained by *glueing* the sheaves $\mathscr{F}_i$ via the isomorphisms φ_{ij}.

Proposition 7.2. [Ha, Exercise II.2.12] and [Sta, Lemma 26.14.1] Let $\{X_i\}$ be a family of locally ringed spaces. For each $i \neq j$, suppose given an open subspace $U_{ij} \subseteq X_i$. Suppose also given for each $i \neq j$ an isomorphism of locally ringed spaces $\varphi_{ij} : U_{ij} \to U_{ji}$ satisfying the cocycle condition:

(1) for each i, $\varphi_{ij} = \varphi_{ji}^{-1}$, and

(2) for each i, j, k, $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$, and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap U_{ik}$.

Then there is a locally ringed space X, together with morphisms $\psi_i : X_i \to X$, such that

(1) ψ_i is an isomorphism of X_i onto an open subspace of X,

(2) the $\psi_i(X_i)$ cover X,

(3) $\psi_i(U_{ij}) = \psi_i(X_i) \cap \psi_j(X_j)$ and

(4) $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij}.

We say that X is obtained by *glueing* the locally ringed space X_i along the isomorphisms φ_{ij}.

Moreover, if each X_i is a scheme, then X is a scheme.

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